| Section | offset | name | nBytes | Comments |
|---|---|---|---|---|
| Game Object | 0 | x coord  world | 4 | |
| | 4 | y coord world | 4 | |
| | 8 | z coord world | 4 | |
| | 12 | Roll Angle | 2 | 32768=pi radians |
| | 14 | Pitch Angle | 2 | |
| | 16 | Yaw Angle | 2 | |
| | 18 | Distance near/far code | 1 | 3=far far away, 2= med far, 1=med near, 0=near (used for Level-of-Detail) |
| | 19 | Class Code | 1 | objhq.class=3, obj5da(plane).class=1, objshard.class = 2, objrv(rescue vehicle). class = 1, bomb.class = 0 |
| | 20 | Distance from camera (squared) | 4 | Probably used to sort objects so they are drawn in the correct order |
| | 24 | object type | 2 | 0=invisible, 0x8=non destructable, 0x800=on hold, 0x104=crashed plane, 0x10=timed object |
| | 26 | rotflag | 1 | 0=no rotation, 2=rotate until med far, 3= rotate until far far away (see distance code) |
| | 27 | hitpoints | 1 | |
| | 28 | Bkpt for d^2 near | 4 | Compare dist to this number to be considered 'near' for level of detail (LOD) |
| | 32 | Bkpt for mednear | 4 | |
| | 36 | Bkpt for med far | 4 | |
| | 40 | Model to use for near | 2 | 16bit memory pointer to model |
| | 42 | model for med near | 2 | 16bit ptr |
| | 44 | model for med far and far | 2 | 16bit ptr |
| | 46 | x velocity | 2 | signed 16bit |
| | 48 | y vel | 2 | |
| | 50 | z vel | 2 | |
| | 52 | timer value | 2 | When gets to zero, timed objs are deleted |
| | 54 | roll angular vel | 2 | |
| | 56 | pitch angular vel | 2 | |
| | 58 | yaw angular vel | 2 | |
| | 60 | dispersion | 2 | Has to do with randomness of path thru space |
| | 62 | damping | 2 | How air resistance slows this down |
| | 64 | Gravity | 2 | gravity effect on this object |
| | 66 | x-size | 2 | collision rectangle x size |
| | 68 | y-size | 2 | collision rectangle y size |
| | 70 | z-size | 2 | collision rect z size |
| | 72 | function ptr for tmr callback | 2 | Ptr into objtimtable in defobjs.asm.  This function is called every frame? |
| | | | | |
| | | | | |
| 3D Model Object | 0 | x coord | 2 | local x coord |
| | 2 | y coord | 2 | |
| | 4 | z coord | 2 | |
| | 6 | Num primitive models | 2 | Models are made of a number of primitives. Each has their own structure.  The array of primitives is tacked onto the end of this structure following this item |
| | 8 | Primitive class | 2 | See cmpjtbl in f3dvec.asm.  0=no draw, 1=frame, 2=point cloud, 3=sphere, 4 = polygon |
| | 10 | Ptr to primitive model data | 2 | Memory ptr to data for primitive.  Different for each type of primitive |
| | 12 | (next primitive class) | 2 | |
| | 14 | next ptr to prim data | 2 | etc etc etc. |

| | | | | |
|---|---|---|---|---|
| | 16 | .... repeats | | |
| | | | | |
| Null PrimitiveData | -- | | | No data for Null primitive |
| | | | | |
| Frame Primitive Data | 0 | Number of wires | 2 | A frame primitive is made up of a number of wires. |
| | 2 | ptr to first wire | 2 | 16 bit ptr to a Wire primitive structure - See wire subprimitive structure below |
| | 4 | ptr to 2nd wire | 2 | |
| | 6 | ptr to 3rd wire ... | 2 | |
| | 8 | ... | | |
| | | | | |
| Point Cloud Primitive Data | 0 | Number of points | 2 | Number of points in the cloud |
| | 2 | Color of points | 2 | If <256, color is constant as specied. Otherwise, this is a 16bit ptr to a color code |
| | 4 | X coord local (pt0) | 2 | |
| | 6 | Y coord Local (pt0) | 2 | |
| | 8 | Z coord Local (pt0) | 2 | |
| | 10 | X coord Local (pt1) | 2 | |
| | 12 | (... and so on ) | 2 | |
| | | | | |
| Sphere Primitive Data | 0 | Number of Spheres | 2 | A sphere primitive is made up of a number of spheres, each can have a different size and color |
| | 2 | X coord Local (sph0) | 2 | |
| | 4 | Y coord local (sph0) | 2 | |
| | 6 | Z coord local (sph0) | 2 | |
| | 8 | Sphere Radius (sph0) | 2 | |
| | 10 | Sphere color (sph0) | 2 | direct color or 16bit ptr to color |
| | 12 | X coord local (sph1) | 2 | |
| | 14 | (...) | | |
| | | | | |
| | | | | |
| Poly Primitive Data | 0 | Number of edges in poly | 2 | Number of sides must be 3 or greater |
| | 2 | Color of polygon | 2 | direct color or 16bit ptr to color |
| | 4 | X coord local (pt0) | 2 | |
| | 6 | Y coord local (pt0) | 2 | |
| | 8 | Z coord local (pt0) | 2 | |
| | 10 | X coord local (pt1) | | |
| | 12 | (...) | | There is one point for each vertex in poly. |
| | | | | |
| Wire Sub-primitive | 0 | Number of segments in wire | 2 | There are n+1 points in a wire that has n segments |
| | 2 | Color of wire | 2 | direct color or ptr to color |
| | 4 | X coord local (seg0) | 2 | |
| | 6 | Y coord local (seg0) | 2 | |
| | 8 | Z coord local (seg0) | 2 | |
| | 10 | X coord local (seg1) | 2 | |
| | 12 | (...) | 2 | |
| | | | | |
| | | | | |
| notes: | | | | |